1^{st} Set of Problems

1. Consider the orthogonal frames $\langle \boldsymbol{x}, \boldsymbol{y} \rangle_1$ and $\langle \boldsymbol{x}, \boldsymbol{y} \rangle_2$ illustrated in Figure 1(a), where the vectors are unitary, $\|\boldsymbol{x}\| = \|\boldsymbol{y}\| = 1$.

- **(a)** Express the coordinates of $\boldsymbol{x}_2$ and $\boldsymbol{y}_2$ in relation to frame one.
- **(b)** Let the coordinates of a vector $\boldsymbol{r}$, expressed in frame two, be $\boldsymbol{r}^2 = \left(r_x^2 \; r_y^2\right)^\intercal$. Express the coordinates of $\boldsymbol{r}$ in frame one, $\boldsymbol{r}^1$.
- **(c)** Express the previous result through a rotation matrix product, $\boldsymbol{r}^1 = \boldsymbol{R}_2^1 \boldsymbol{r}^2$.

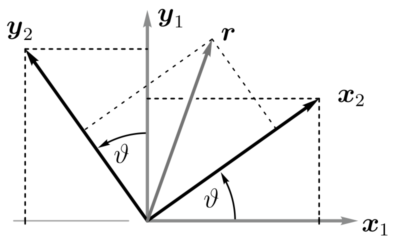

Figure 1: Rotation matrix as a change in the vector's frame representation.

2. Consider the elementary rotations illustrated in Figure 2.

- **(a)** Following the method in Problem 1, write the rotation matrices $\boldsymbol{R}_2^1$.
- **(b)** Write the rotation matrices $\boldsymbol{R}_1^2$ by first showing that $\boldsymbol{R}_1^2 = \left(\boldsymbol{R}_2^1\right)^\intercal$.

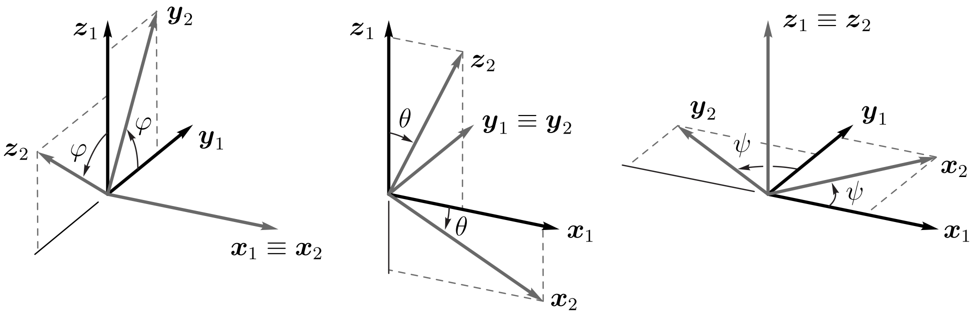

Figure 2: Elementary rotations.

3. Employ two consecutive rotations to prove the following angle sum identities,

$$\cos(\alpha \pm \beta) = \cos(\alpha)\,\cos(\beta) \mp \sin(\alpha)\,\sin(\beta)\ ,$$
$$\sin(\alpha \pm \beta) = \sin(\alpha)\,\cos(\beta) \pm \cos(\alpha)\,\sin(\beta)\ .$$

4. Employ the distance formula to prove the law of cosines, given by

$$c^2 = a^2 + b^2 - 2\,a\,b\cos(\vartheta)\ ,$$

where a, b and c are the side lengths of a triangle and ϑ the inside angle between a and b.